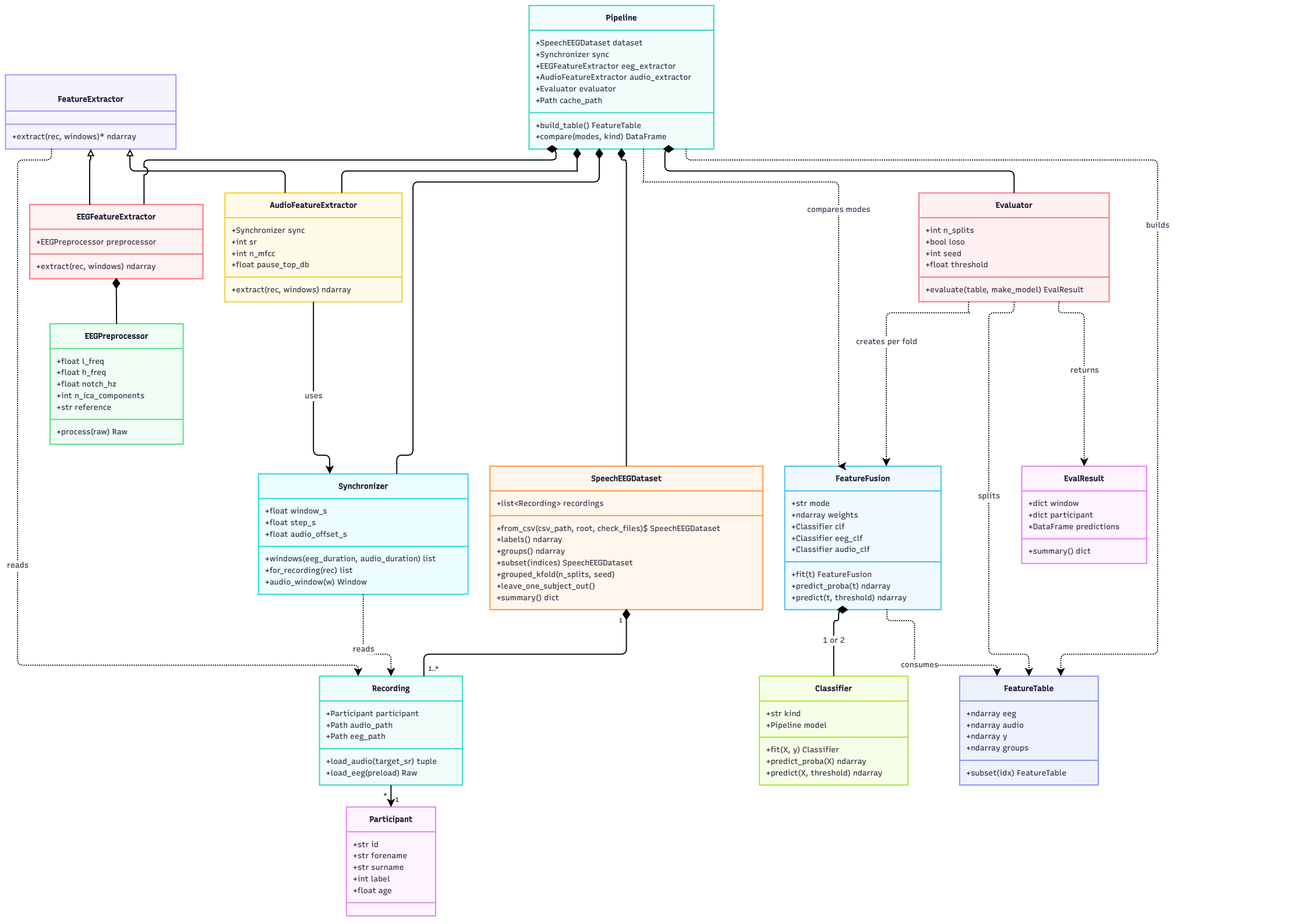

Pipeline
+SpeechEEGDataset dataset
+Synchronizer sync
+EEGFeatureExtractor eeg_extractor
+AudioFeatureExtractor audio_extractor
+Evaluator evaluator
+Path cache_path
+build_table() FeatureTable
+compare(modes, kind) DataFrame
FeatureExtractor
+extract(rec, windows)* ndarray
EEGFeatureExtractor
+EEGPreprocessor preprocessor
+extract(rec, windows) ndarray
AudioFeatureExtractor
+Synchronizer sync
+int sr
+int n_mfcc
+float pause_top_db
+extract(rec, windows) ndarray
Evaluator
+int n_splits
+bool loso
+int seed
+float threshold
+evaluate(table, make_model) EvalResult
EEGPreprocessor
+float l_freq
+float h_freq
+float notch_hz
+int n_ica_components
+str reference
+process(raw) Raw
Synchronizer
+float window_s
+float step_s
+float audio_offset_s
+windows(eeg_duration, audio_duration) list
+for_recording(rec) list
+audio_window(w) Window
SpeechEEGDataset
+list<Recording> recordings
+from_csv(csv_path, root, check_files)$ SpeechEEGDataset
+labels() ndarray
+groups() ndarray
+subset(indices) SpeechEEGDataset
+grouped_kfold(n_splits, seed)
+leave_one_subject_out()
+summary() dict
FeatureFusion
+str mode
+ndarray weights
+Classifier clf
+Classifier eeg_clf
+Classifier audio_clf
+fit(t) FeatureFusion
+predict_proba(t) ndarray
+predict(t, threshold) ndarray
EvalResult
+dict window
+dict participant
+DataFrame predictions
+summary() dict
Recording
+Participant participant
+Path audio_path
+Path eeg_path
+load_audio(target_sr) tuple
+load_eeg(preload) Raw
Classifier
+str kind
+Pipeline model
+fit(X, y) Classifier
+predict_proba(X) ndarray
+predict(X, threshold) ndarray
FeatureTable
+ndarray eeg
+ndarray audio
+ndarray y
+ndarray groups
+subset(idx) FeatureTable
Participant
+str id
+str forename
+str surname
+int label
+float age
compares modes
builds
uses
creates per fold
returns
splits
reads
reads
1
1..*
1 or 2
consumes
*
1